Signalling with objects

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Conventional signalling systems, such as Q.931 and Q.2931, have grown from simple techniques for creating a single connection between two parties. However, with B-ISDN, it will be desirable to be able to construct multiparty, multiconnection calls for multimedia services. This paper highlights the work done by the RACE MAGIC project to develop a new object-based signalling protocol to meet these requirements, and describes how this new protocol overcomes the shortcomings of conventional signalling.

1. Introduction

Conventional public switched telephone network (PSTN) and narrowband ISDN (N-ISDN) signalling systems have evolved from the need to support the two-party telephony service. However, the development of broadband ISDN (B-ISDN), with its associated advanced switching and control capabilities, results in a network that can support services that are far more advanced than telephony. In particular, using asynchronous transfer mode (ATM) permits multiple connections to be established over the same transmission pipe, and advanced switching can support more than two parties in a call, while in the future there will be many more communications services on offer than just telephony.

Thus there is the requirement for B-ISDN to signal multiparty and multiconnection calls across the network. This is a very complex task, especially once one considers that some terminals may not be compatible with the others, lack facilities for some media types in use, and require special resources such as conference bridges within the network; parties may also join/leave the call.

The network technology already exists to perform all of the above. The difficulty lies in communicating the requirements between terminals and the network. It can therefore be seen that signalling is the key enabler. Inadequate signalling can prevent users fully utilising the capabilities of the network.

At the start of the 1990s the existing signalling protocols, such as the initial releases of Q.2931 [1] and B-ISUP (broadband ISDN user part) [2], were inadequate for such advanced services, and effort was needed to develop the signalling protocols that would be required in the future. Consequently the European Commission established the RACE 2044 (MAGIC) project to undertake pre-competitive research into exactly these issues [3]. A number of network operators, equipment providers, and universities, with BT acting as prime contractor, collaborated between 1992-94 to develop advanced user/network interface (UNI) and network/node interface (NNI) signalling protocols for the B-ISDN. Their remit was to look to the future and to provide a longer term view to the international standards bodies for signalling.

The following companies contributed to the MAGIC project — BT (UK), ATEA (B), AT&T (NL), University of Twente (NL), Alcatel Bell (B), Alcatel CIT (F), CSELT (I), Ericsson Telecommunicazione (I), NTUA (GR), Telekom FTZ (D), Technische Universität Ilmenau (D), PTT Research (NL), Bellcore (USA), Italtel (I), RACE Industrial Consortium (RIC).

This paper discusses the signalling system developed within this project and describes how it can be used to support complex services. Knight and Law [4] describe how some of the techniques developed within the MAGIC project are now being incorporated by the ITU-T into the B-ISDN standards.

2. Limitations of the current approach

The access signalling protocol currently being developed for broadband ISDN is Q.2931 (also called Digital Subscriber Signalling System No 2 — DSS2) [1]. The early releases of this were essentially a development of the narrowband ISDN's signalling protocol Q.931 (DSS1) [5] with alterations to accommodate the ATM transport layers. Unfortunately this evolution means that Q.2931 inherits many of the limitations of Q.931 and its ancestors. When applied to multi-party, multiconnection calls, extending Q.2931 becomes unwieldy and complicated, essentially requiring the signalling state machines to be modified for each new type of service supported. This is due in part to the architecture behind Q.2931 deriving directly from early tel-ecommuni-cations signalling protocols designed purely for telephony (see Fig 1).

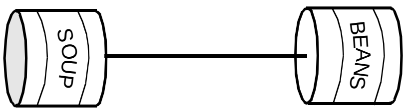

Fig 1 Underlying architecture of Q.2931.

Q.2931 is designed for two-party communication with a single bearer connection between the parties. When the call is established, the calling party controls all of the attributes of the call according to its model of how the call should exist through the network. However, as services become more advanced, such an overall view of the call becomes more difficult; in fact, in some services it may even be difficult to determine who is the caller and who is the called party. For example, consider a hypothetical travel agency service [6].

A customer uses their videophone to enquire about a holiday. During the conversation the travel agent contacts a multimedia video library to select a movie clip for the holiday in question. The travel agent then instructs the videolibrary to play the movie clip for the holiday directly to the customer. At this point the call model has the appearance of the arrangement in Fig 2.

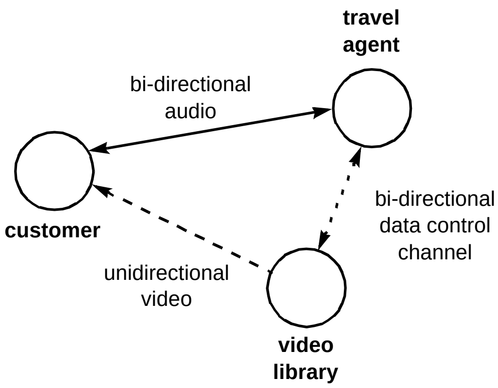

Fig 2 Travel agency service.

In the above service there is a mix of parties creating connections, not just the original caller. Some of these connections are bi-directional and some unidirectional. Also, it can be seen that not all parties participate in all connections. For instance, the customer need never be aware of the data link between the travel agent and the video library. Thus in this case the original caller need not have a view of everything that exists during the lifetime of the call. It can also be seen from Fig 2 that the travel agent is setting up the video connection to the customer without using the video itself. This gives an example of the control flows for the call being separate from the signalling used to establish the bearer connections. This requires separation of call and bearer control.

Another issue that has to be dealt with is interworking between the different codecs used to encode information. It is pointless for the video library to establish an MPEG [7] video connection to a customer whose equipment only supports H.261 [8] video. What is really required is a mechanism to invoke network- or terminal-based interworking to deal with such mismatches, without failing the call.

The travel agency service is a fairly complex service. It is impossible for the network to directly build in support for every such service that may come about. Instead what is required is a signalling protocol that can accommodate flexibly any such service requirements without needing to know the overall architecture of the service. This suggests that it is necessary for the signalling protocol to permit applications to manipulate connections, support parties undertaking different service roles and using terminals of differing capabilities, effectively treating them as a set of resources, or objects, to be mixed and matched as necessary. Also, with the high degree of complexity of such services, it is necessary for the signalling protocol to permit these manipulations to take place in a controlled way so that the network never ends up in an indeterminate state.

3. MAGIC signalling model

Existing signalling protocols have a number of weaknesses when applied to the complex multi-party, multimedia world of broadband ISDN. Therefore the MAGIC project developed a number of new signalling concepts to counteract these weaknesses and to build the basis for a flexible signalling system for broadband. This section describes the main concepts introduced by the project [8].

3.1 Separation of protocols

It was a basic assumption of MAGIC that the network would support separation between call and bearer control, at least within the internodal (NNI) protocols. It was determined within the MAGIC project that call/bearer protocol separation was not necessary at the user/network interface (UNI) and thus a monolithic protocol was devised for this.

Effectively call control (CC) is an end-to-end negotiation protocol that permits users to determine the call configuration they desire. Once determined, bearer control (BC) is used to establish the required bearers throughout the network. With this separation, not all nodes need implement the call-control signalling stack, e.g. call control could be included in the local exchanges (LEX) and omitted from the transit exchanges (TEX) (Fig 3). To provide an evolutionary

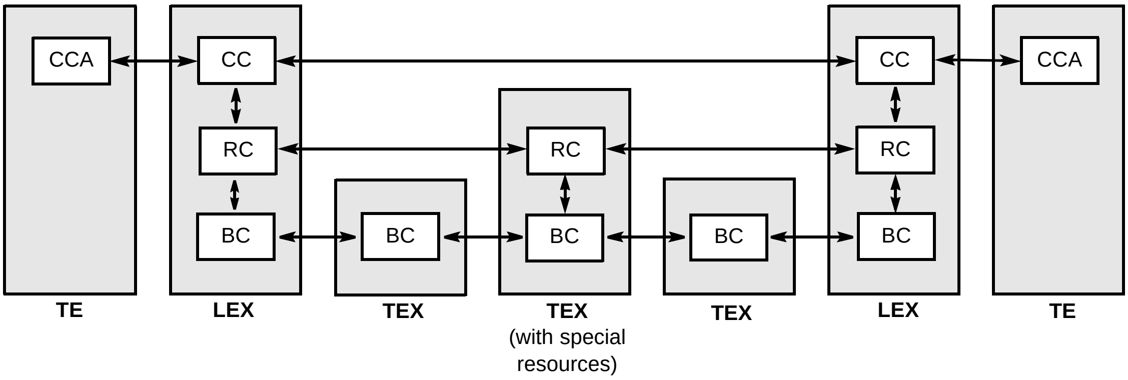

Fig 3 CC, BC, and RC signalling stacks

path from existing networks, B-ISUP was employed as the bearer control protocol.

When the MAGIC project investigated the complexity of multimedia calls, and in particular the problems involved in allocating special resources such as interworking units, conference bridges, etc, it was found necessary to develop an additional protocol, resource control (RC), to manage these special devices within the network. Call control could not be used for this purpose as resource management would violate the principle of call control being an end-to-end signalling protocol.

At the UNI a monolithic protocol was devised to perform the call control and bearer control functions. This entity is termed the call control agent (CCA).

3.2 *Local views*

Traditionally the terminal creating a call signals all the information related to that call to all participants in the call. Thus it could be said that there exists a ‘global view’ for this call, i.e. all of the relative attributes for the call can be described as a single entity.

With the complexity of broadband ISDN, where different parties in a multi-party call are using different communications flows, such a ‘global view’ can become unwieldy. Thus in MAGIC the concept of ‘local views’ was adopted.

A ‘local view’ is the view a particular terminal has of the call. Basically it only describes the parts of the call with which that terminal has an association, e.g. the remote parties with which the terminal is communicating, or the connections that the terminal has established. If objects are not of relevance to a terminal, e.g. a connection between two remote parties only, they need not be represented in that the terminal’s ‘local view’.

Each terminal has a ‘local view’ that describes that terminal’s view of the service. This will be different to the ‘local views’ of other terminals in the call. With this approach there is no single ‘global view’ of the call, it is only when all the individual ‘local views’ are brought together that the entire call can be described (see Fig 4).

3.3 *Abstract services*

With the increased communications options supported by broadband ISDN, interworking between terminals using different encoding standards for media sources will become increasingly difficult. For example, two terminals may wish to establish a videotelephony call, but one terminal uses H.261 video standards, and the other uses MPEG. If the network can offer an H.261/MPEG converter, in theory this call could be connected. The problem lies in signalling this between the terminals and network.

MAGIC introduced the concept of ‘abstract service’. This describes encoding independent values for the communications media, i.e. video in the example above. For a videotelephony call, both terminals must accept the same video abstract service, but may signal different encoding parameters for use within their own ‘local view’ of the call. The fact that they are using encoding types that are different

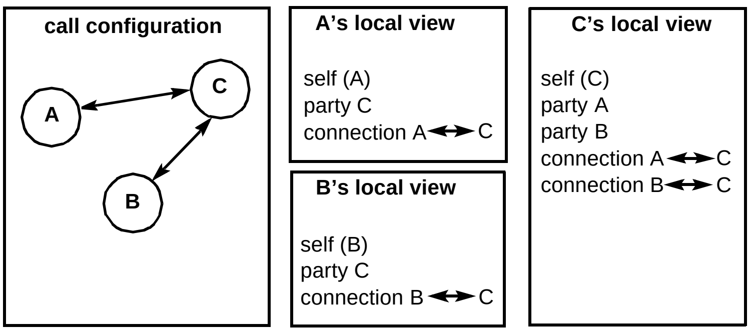

Fig 4 ‘Local view’ concept.

from the remote terminal is irrelevant as the network deals with the required translation, thus permitting incompatible terminals to interact with the service. The abstract service therefore enables the global attributes of the communication for that media to be described, while the encoding types remain local to each terminal.

3.4 MAGIC call control objects

The MAGIC signalling protocols essentially establish calls by manipulating a set of signalling objects. This object-oriented approach permits complex services to be built from a reusable set of resources. This provides a powerful signalling capability based on a set of simple objects and operations on these objects.

Figure 5 shows the signalling object model and Table 1 lists the object definitions.

Signalling for broadband calls is essentially the process of creating, modifying and deleting these objects in a terminal's local view. For example, to create a two-party telephone call, a terminal would have to create the following objects:

- party set (specifying telephony service),
- confirmed party (called party),
- abstract service (for audio),
- abstract service module (for the audio codec used by the caller),
- upstream/downstream mapping elements (suitable for audio codec),
- confirmed party service link (linking the remote party to the audio service).

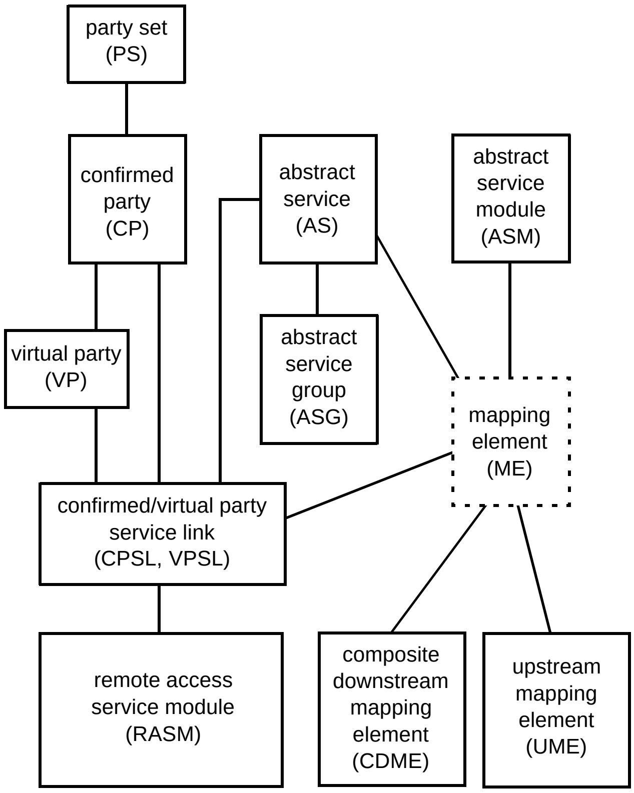

Fig 5 Signalling object model.

The remote party will have a similar set of objects created except that the abstract service module and mapping elements may be different to the caller (if interworking is

Table 1 Object definitions.

Object	Description
Party Set (PS)	This defines an association between a set of parties, and thus essentially defines a call, and the service being supported by this call. All parties in a call must share the same Party Set object.
Confirmed Party (CP)	A Confirmed Party is a party that is diretly involved with the negotiation of bringing a party into the party set. For example, you add a party to a call by creating a Confirmed Party object for it. This party will similarly have a Confirmed Party object created representing the initiator.
Virtual Party (VP)	A Virtual Party is a party who is not being added to the Party Set for a particular local view. Thus if a call exists between parties A and B, and A adds a new party C to the call, then C will appear as a Virtual Party in B's local view.
Abstract Service (AS)	This is an abstract 'global' representation of a media communication type, or service module as it is termed in MAGIC. For example, this could be video or audio.
Confirmed/Virtual Party Service Links (CPSL/VPSL)	A Party Service Link associates a particular Abstract Service with the relevant party. Thus if a remote Confirmed Party is to have audio and video connections, then two Confirmed Party Service Links need to be created to link this party with the audio and video Abstract Service objects.
Abstract Service Group (ASG)	The Abstract Service Group permits a number of Abstract Services to be synchronised in some way, perhaps to ensure the audio and video streams are in lip-synchronisation with each other.
Abstract Service Module (ASM)	An Abstract Service Module encapsulates the user end-point of a bearer connection. It thus represents the attributes for that connection, such as ATM channel attributes and the encoding standards used.
Remote Access Service Module (RASM)	This is a virtual object that provides information of the Access Service Module used by a remote party. If a Party Service Link does not have a RASM then it indicates that that party has no bearer for this Abstract Service set-up. This object also permits 3rd party call set-up (where a party sets up a connection between remote parties only).
Mapping Element (ME) Composite Downstream Mapping Element (CDME) Upstream Mapping Element (UME)	A Mapping Element describes a unidirectional flow of information between an Abstract Service and an Abstract Service Module. It can effectively be thought of as a representation of an ATM connection. Two mapping elements are normally needed for a connection, one for the downstream link, and one for the upstream link.

taking place within the network). Once these objects are created, the telephony call will exist in the network.

3.5 Atomic action protocol

Complex broadband services will require complex sets of information to be signalled. For example, multiple parties may be added to calls, each using different sets of connections, while the connections to other parties are being modified. This results in complex interactions that need a mechanism to ensure the system ends up in a stable and correct state.

MAGIC employed an atomic action protocol (AAP) to signal these changes. Atomic actions are operations that act on entities that must succeed or fail as a unit. This protocol is based open the ITU X.851 commitment, concurrency, and recovery (CCR) protocol [9].

When an entity wishes to change the state of the call, e.g. to add parties to a call, it starts an atomic action that requests the other entities to accept or reject this change (see Fig 6). If they all accept the change, then the initiating entity will tell them to commit to the change and all entities will update themselves to the new values. If any one of the entities fails to accept the change, then the atomic action will be 'rolled back' and all objects will return to their initial state. Thus, in the previous example, all parties being added to the call would have to accept the request, otherwise none would be added.

When deciding whether an atomic action request can be accepted, an entity may have to cascade this request to one or more other entities. These other entities may further cascade the entities, building up an atomic action tree. In this tree, unless all of the terminating entities accept the atomic action, then all of the atomic actions will fail. An example of this is adding parties to a call. A terminal will signal the creation of new party objects in an atomic action to its exchange. This may then cascade two further atomic actions, one to create each party object to the exchanges to which these parties are connected, and secondly to the remote parties themselves. Both parties must accept this creation for the originator to receive the READY indication (see Fig 7).

In the example in Fig 7, A wishes to create a call to B and C. A initiates an atomic action to add B and C to the call. These cascade two further atomic actions for B and C through the local exchanges. B accepts this with a READY, while C refuses it with a ROLLBACK. Since at LEX1 one of the child atomic actions has failed, the entire atomic

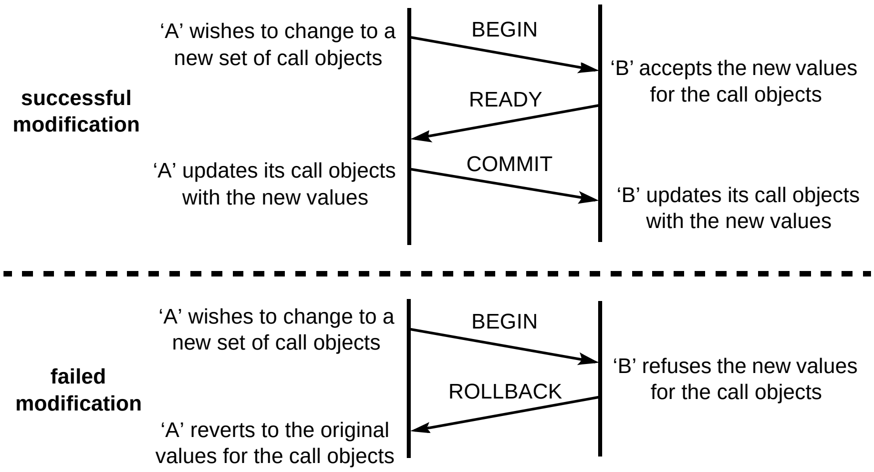

Fig 6 Atomic actions.

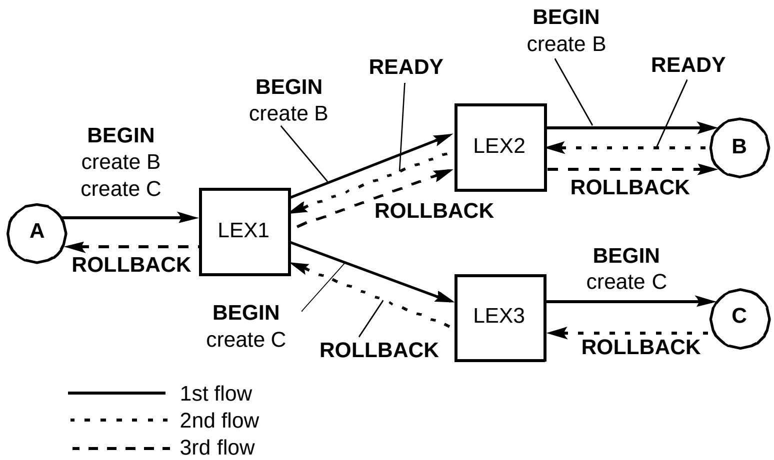

Fig 7 Atomic action tree example.

action tree fails. Therefore LEX1 sends a ROLLBACK to all remaining atomic actions in the tree.

A further refinement of the atomic action protocol is the ability to nest additional atomic actions within a parent one. The child operations can be labelled as mandatory or optional — mandatory meaning that the child must succeed for the parent to succeed, optional meaning that the child may fail without affecting the success of the parent. However, if the parent operation fails, then all child operations fail.

This nesting of atomic operations provides the modularity of the atomic action protocol, while permitting optional operations to be indicated. For example, in a videotelephony call, one nested atomic action could be used to establish the audio connection, and another optional atomic action to establish the video. A terminal may accept the overall atomic action with the audio while refusing the video (perhaps it does not support video).

4. MAGIC signalling example (videoconference call)

This section describes an example of setting up a videoconference call using the MAGIC signalling protocol.

In Fig 8, terminal TE1 will establish a videoconference call to two other terminals — TE2 and TE3. However, only TE2 will accept both the video and audio connections, TE3 will accept the audio only.

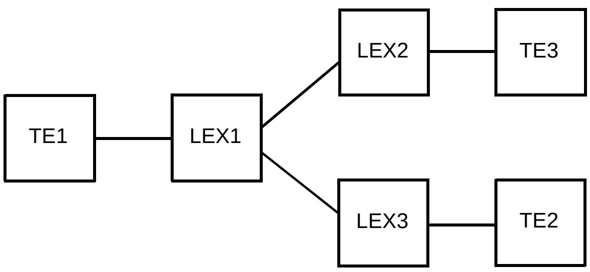

Fig 8 Network configuration for videoconference example.

Figure 9 shows the signalling flows that occur across the network. Since, in this example, the bridging of the audio and video streams is performed wholly within LEX1, no resource control signalling flows will exist.

Table 2 gives details on the meaning of the flows in Fig 9, and examples of the contents of key messages. The message element contents have been simplified to give just the basic meaning.

By the end of this call set-up, each party will be participating in the service and have a set of established call objects appropriate to their own local view. For example,

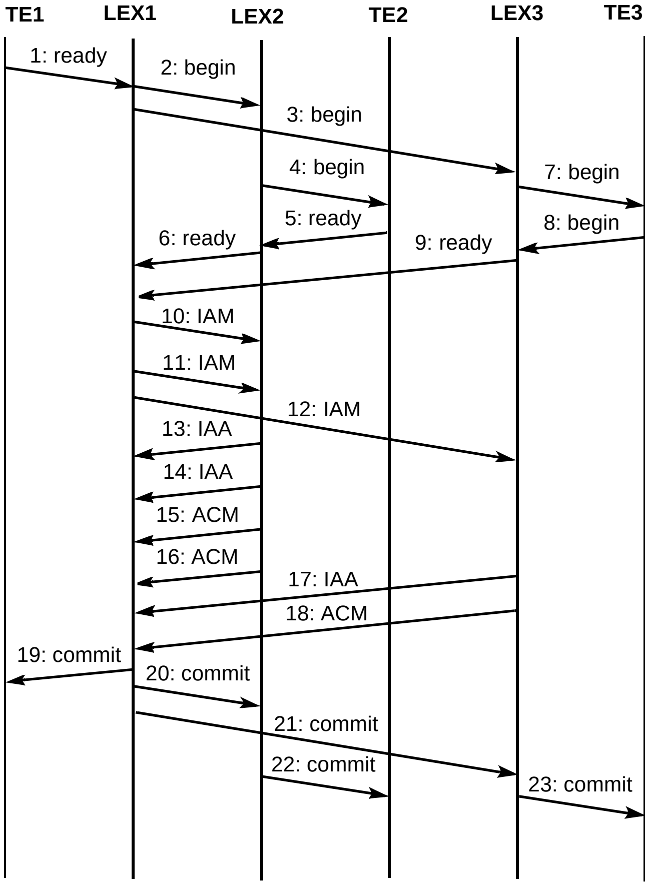

Fig 9 Flows for setting up a videoconference call.

Fig 10 shows the set of call objects that TE1 will have in its local view.

5. Discussion and future

Part of the work performed by the MAGIC protocol was the building of a software simulator to test the protocol s being developed. As part of this work, experiments were performed in supporting services with both an extended version of Q.2931 and the UNI atomic action protocol. The latter was found to be much easier to implement, only requiring simple signalling actions to be performed in the correct order. Using Q.2931 essentially required the signalling sub-systems to be specifically tailored for each individual service, in order to communicate all the complex requirements of the service. Thus it was concluded that the object approach to signalling, as used by MAGIC, was far more flexible and simpler to use. This is, however, a radical departure from conventional signalling protocols and the the object-based approach has only recently been investigated by the standards bodies [4].

For the services examined by the MAGIC project the object-based signalling model and the atomic action protocol was found to meet all of the requirements of these services. In particular, it was able to signal the following requirements that would be very difficult to achieve with conventional protocols:

Table 2 Set-up flow message descriptions.

Message	Description	Example message contents

1: Ready	TE1 signals over the access interface that it wishes to establish a videoconference call, with audio and video, to TE2 and TE3.
	
	This signalling message starts an atomic action to create the objects in TE1's local view of the call, that is:
	• the service
	• two confirmed parties
	• audio and video abstract services
	• the coding and bearers TE1 will use for the audio and video.
		`1:READY`
		`create(PS:Videoconference)`
		`create(CP:TE2)`
		`create(CP:TE3)`
		`create(AS:audio)`
		`create(CPSL:TE2/audio)`
		`create(CPSL:TE3/audio)`
		`create(ASM:A-law)`
		`create(UM:64kbit/s)`
		`create(CSDM:64kbit/s)`
		`create(AS:video)`
		`create(CPSL:TE2/video)`
		`create(CPSL:TE3/video)`
		`create(ASM:H.261)`
		`create(UME:128kbit/s)`
		`create(CDME:128kbit/s)`
		`create(ASG:audio,video,sync)`

2: Begin 3: Begin	The originating LEX assigns video and audio bridge resources internally. It then continues the call set-up process by contacting the destination LEXs setting up similar sets of call objects to those in message 1.	

4: Begin
7: Begin
	The LEXs signal the call attempt to the called TEs.
	
	(Note how the objects relate to the local view of the called TE, e.g. the other called party is seen as a virtual party in this view.)
		`4:BEGIN`
		`create(PS:Videoconference)`
		`create(CP:TE1)`
		`create(VP:TE3)`
		`create(AS:audio)`
		`create(CPSL:TE1/audio)`
		`create(VPSL:TE3/audio)`
		`create(AS:video)`
		`create(CPSL:TE1/video)`
		`create(VPSL:TE3/video)`
		`create(RASM:TE1,A-law)`
		`create(RASM:TE1,H.261)`
		`create(ASG:audio,video,sync)`

5: Ready
6: Ready
	TE2 accepts the call. This TE wishes to use both audio and video and thus selects the encoding schemes that this terminal wishes to use and the mapping elements necessary for the bearer connections. The acceptance is passed back to the originating LEX.
	
	(Note: most of the call objects created by message 4 are implicitly accepted by this message and are not signalled explicitly. Only the newly created objects need be signalled).
		`5:READY`
		`create(ASM:A-law)`
		`create(UM:64kbit/s)`
		`create(CDSM:64kbit/s)`
		`create(ASM:H.261)`
		`create(UME:128kbit/s)`
		`create(CDME: 128kbit/s)`
		
		`6:READY`
		`create(RASM:TE2,A-law)`
		`create(RASM:TE2,H.261)`

8: Ready
9: Ready
	Similarly TE3 accepts the call, but only wishes to use audio. Thus it only creates an ASM and mapping elements for the audio stream (note that it also selects a different audio encoding type than that of the other terminals).
	
	Once the originating LEX has received both confirmations from the destination LEXs, and has ascertained that it has the necessary A-law↔μ-law interworking unit, call control processing stops and bearer control takes place to set up the network bearers between LEXs.
		`8:READY`
		`create(ASM:μ-law)`
		`create(UME:64kbit/s)`
		`create(CDME:64kbit/s)`
		
		`9:READY`
		`create(RASM:TE3,μ-law)`

10: IAM 11: IAM 12: IAM	Normal B-ISUP initial address messages are sent to set up the bearer connections. 10: audio bearer to TE2 11: video bearer to TE2 12: audio bearer to TE3	
13: IAA 14: IAA 15: ACM 16: ACM 17: IAA 18: ICM	Standard B-ISUP flows. *When LEX 1 has received all three Address Complete Messages (ACM), the bearers have been established. LEX 1 can now complete call control and finish the set-up phase of the call.*	

19: Commit
20: Commit
21: Commit
22: Commit
	Call control will now complete by confirming all of the pending atomic actions. The created call objects are now also confirmed. On the accesses, the virtual path and virtual channel identifiers (VPIs/VCIs) assigned to the different ATM connections are signalled to the terminals.	`19:COMMIT`
		`create(RASM:TE2,A-law)`
		`create(RASM:TE2,H.261)`
		`create(RASM:TE3,μ-law)`
		`create(ASM: A-law,VPI,VCI)`
		`create(ASM: H.261,VPI,VCI)`

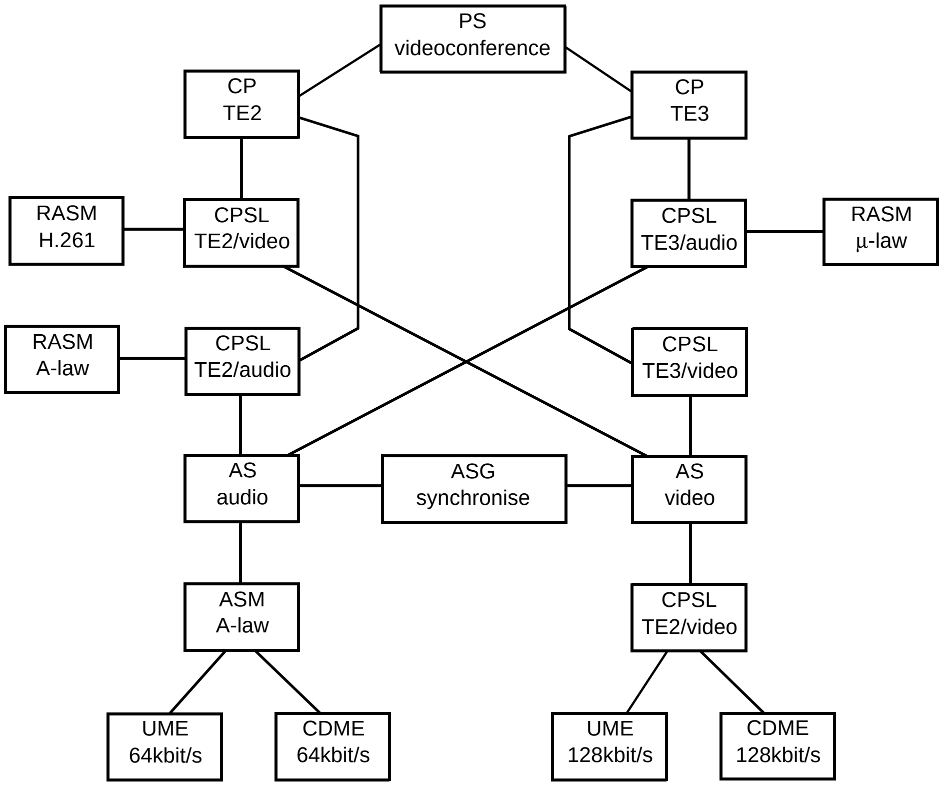

Fig 10 TE1's local view of the videoconference call.

- support multi-party calls,
- add/drop parties during the call,
- permit parties to undertake different roles (i.e. called parties to add/remove bearers and other parties),
- provide a mechanism to facilitate interworking of codecs,
- support advanced services like remote control (party A sets up a bearer between parties B and C) and a user-defined service (a party connects to other parties using connections as required, and not to any formal service definition).

The MAGIC work does not necessarily remove the complexity of application control for such services (such as working out the necessary billing mechanisms). What it did do was remove much of the complexity from the signalling protocol. Signalling is there to communicate requirements across the network, not restrict them.

The major weakness of the MAGIC work was its failure to address the requirements of data networks, such as TCP/IP, to utilise the capabilities of the intelligent network (IN), and to address fixed/mobile network convergence. Data networks were not as important when the MAGIC project existed as they are today, and further investigation is needed to assess how public networks should support such an important service. As for the IN, there appears to be much in common between the services provided by the MAGIC resource control protocol and the IN. It can therefore be envisaged that the resource control functions will be provided instead by IN capabilities.

Finally, since mobile networks did not have the same penetration in 1992 as they do today, no work was performed in MAGIC to support them. However, many of the MAGIC concepts, such as call/bearer control separation could facilitate fixed/mobile convergence.

6. Conclusions

Current signalling protocols for broadband ISDN are essentially an evolution of the simple protocols developed to support the two-party, single-connection service of telephony. Current approaches do not adapt to the multi-party multiconnection services envisaged for broadband. Therefore MAGIC developed a new protocol for such advanced services. It was designed around the concept of communicating simple objects using an atomic action protocol to build associations between functional entities. By using such a signalling system, it was possible to set up calls for advanced broadband services in a relatively simple way, demonstrating that this approach was indeed a good way of flexibly supporting the advanced services of tomorrow.

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